

Federated learning model with dynamic scoring-based client selection for diabetes diagnosis

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ABSTRACT

Federated learning (FL) is a privacy-preserving paradigm in distributed machine learning that enables clients to collaboratively train models without sharing their raw data. However, the variety of client data, device setups, and network conditions provides serious difficulties for FL systems. Random client sampling in such environments often leads to suboptimal outcomes, including lower model accuracy, slower convergence rates, and reduced fairness. To address these issues, this study proposes a dynamic client selection mechanism based on a scoring system that evaluates clients based on three key parameters: accuracy, loss, and execution time. We propose a scoring-based framework for adaptive client selection in federated learning (FL) and implement it in an ML-driven diabetes detection system. Evaluations with 200 communication rounds demonstrate improved global and local model performance, faster convergence, and optimized resource utilization. The framework dynamically selects clients, improving execution efficiency and addressing key FL challenges, including data heterogeneity, fairness, communication overhead, and privacy. Our findings highlight its potential for scalable and efficient FL in healthcare applications while paving the way for future advancements in adaptive client selection.

1. Introduction

Federated Learning (FL) is an emerging paradigm in machine learning that allows multiple devices (often edge devices) to collaboratively train a model without sharing raw data. Instead, these devices only share updates to model parameters, preserving user privacy and data security. The future of federated learning promises to significantly impact the fields of AI and ML by improving performance, efficiency, and convergence. Enabling decentralized model training reduces the need for large-scale data sharing, preserving privacy and security. As computational resources improve, federated learning will achieve faster model convergence, making it more practical for real-world applications. Enhanced communication protocols will minimize latency and reduce energy consumption. Also, hybrid federated learning models that integrate edge and cloud computing will further boost scalability. Innovations in model aggregation will enable better generalization and robustness. With improved algorithms, federated learning will

enhance its ability to handle heterogeneous data. These advancements will make federated learning a key enabler for healthcare, finance, and IoT sectors. Ultimately, federated learning will contribute to a more collaborative and privacy-preserving AI ecosystem. The future of federated learning is poised to redefine how machine learning models are trained, making them more efficient, secure, and personalized. As advancements continue in optimization, communication, and privacy techniques, federated learning will be essential in domains such as healthcare, finance, and autonomous systems, where data privacy and real-time learning are crucial.

Federated Learning (FL) systems depend heavily on client selection, which significantly affects the overall effectiveness, efficiency, and fairness of the training process. The efficacy of the federated learning framework is shaped by the process of choosing which clients to include in each training cycle. This approach directly impacts the

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model's quality, the efficient use of computational resources, and the fair participation of varied clients. FL has emerged as a transformative paradigm for collaborative model training across distributed clients, enabling the preservation of data privacy by keeping sensitive data localized. However, the heterogeneity of clients introduces complex challenges that make selecting an optimal subset of participants for each training round highly intricate. The diversity among clients encompasses disparities in computational resources, data quality, and network conditions. Such variations can significantly affect the training dynamics, as computationally limited clients may introduce delays, while clients with low-quality or non-representative data may degrade model performance. Furthermore, non-uniform network connectivity can lead to inconsistencies in communication, impacting system efficiency and convergence speed. Effective client selection strategies are, therefore, essential for navigating these challenges. They must account for clients' diverse characteristics, including variations in data distribution (IID vs. non-IID scenarios), resource availability, and communication constraints. By carefully balancing these factors, client selection mechanisms can optimize model performance, accelerate convergence, and enhance the overall efficiency of the FL framework. In addition to performance and efficiency, fairness has become a critical consideration in client selection. Ensuring equitable participation among underrepresented clients or datasets is crucial for building robust and unbiased models. Advanced client selection methodologies incorporate adaptive mechanisms to address the multifaceted challenges inherent in federated learning. These techniques ensure that the chosen subset of clients significantly enhances global learning while upholding system reliability, optimizing resource utilization, and supporting scalability in diverse and dynamic environments.

Client selection is a fundamental FL component, significantly influencing performance, efficiency, and fairness. Effective selection mechanisms must address diverse factors, including data distribution, client heterogeneity, privacy constraints, and resource limitations, each playing a critical role in shaping the effectiveness of FL models. Data distribution across clients is a key determinant of model performance. Techniques like FedEntOpt maximize the entropy of the global label distribution, ensuring the model is trained on diverse data labels, which enhances accuracy and robustness, particularly in scenarios with low client participation [1]. Similarly, RICA uses data size and entropy as selection criteria to improve accuracy and resilience against malicious clients [2]. Client heterogeneity, encompassing variations in data quality, computational capacity, and hardware capabilities, requires balanced selection to maintain fairness and efficiency. Algorithms like SUBTRUNC and UNIONFL leverage submodular function maximization to ensure equitable model performance across clients, which is especially critical in sensitive domains such as healthcare [3]. In resource-constrained environments, approaches like the Grey-Wolf Optimizer account for multiple attributes, including energy consumption and network latency, to optimize client selection, promoting fairness and reliability [4]. Differential privacy further complicates selection, as privacy requirements vary among clients. The DPFL-BCS algorithm addresses this by optimizing client selection to balance privacy constraints with model utility, minimizing loss while maintaining security [5]. Additionally, RICA incorporates mechanisms to identify and exclude malicious clients, bolstering security and model accuracy [2]. Emerging methods like over-the-air federated learning (OTA-FL) introduce unique energy consumption and network latency challenges. The Grey-Wolf Optimizer approach effectively manages these constraints, reducing convergence time and improving energy efficiency while sustaining high model performance [4]; despite advancements, challenges persist, particularly in balancing trade-offs between privacy, performance, and fairness. Future research should explore adaptive and dynamic client selection mechanisms capable of responding to real-time conditions, further enhancing the outcomes and scalability of federated learning systems.

Federated Learning (FL) employs a distributed machine learning framework in which a central server aggregates models while individual clients update the model locally. This approach enables clients to maintain control over their data by transmitting only the updated model parameters to the central server, safeguarding privacy by preventing the disclosure of raw data. However, FL faces several critical challenges. Privacy is a primary concern, as local data ownership provides baseline protection but does not eliminate vulnerabilities to adversarial attacks, such as backdoor attacks [6], model poisoning [7], and data poisoning [8]. Communication costs represent another significant challenge due to the frequent uploading and downloading of model updates. Improving communication efficiency is essential [9–11], as the centralized architecture places a heavy workload on the central server, prompting the exploration of decentralized architectures to mitigate these issues [12]. Because client data is often non-independent and identically distributed (non-IID), statistical heterogeneity poses an extra hurdle in federated learning. This client-to-client variation may lead to less than ideal global model performance. To mitigate this issue, several advanced model aggregation techniques have been proposed. These include adaptive weighting strategies [13,14], attentive aggregation methods [15], regularization approaches [16], clustering techniques [17], and Bayesian aggregation methods [18], all of which aim to enhance the robustness and effectiveness of model aggregation in the presence of heterogeneous client data. These strategies aim to refine the weighting of client contributions based on characteristics beyond just data quantity [19]. Moreover, fairness in aggregation has become a focus, emphasizing the inclusion of underrepresented clients and classes to enhance the global model's equity and representativeness. These advancements are shaping FL to handle better privacy, efficiency, and heterogeneity in decentralized machine learning systems.

Our key contributions are as follows:

- Proposed a novel dynamic client selection mechanism based on a scoring system, utilizing accuracy, loss, and execution time as scoring parameters, addressing the heterogeneity of clients in federated learning (FL).
- Developed and implemented an FL-based diabetes detection system incorporating the proposed scoring-based client selection, optimizing resource utilization and enhancing model performance, convergence speed, and fairness.
- Conducted extensive experiments over 200 communication rounds, offering critical insights into global accuracy and loss trends, local device performance, scoring patterns, and execution times.
- This work incorporated federated learning for ML-driven diabetes detection by tackling statistical heterogeneity, resource constraints, and scalability.

This paper's remaining sections are organized as follows: Section 2 gives a thorough assessment of the literature state. Section 3 explores the foundational principles and background of federated learning (FL) within the healthcare industry, highlighting its distinctive characteristics and benefits. Section 4 details the client selection process in federated learning, including the scoring criteria and the proposed dynamic client selection methodology based on a scoring system. The findings and discussion are presented in full in Section 5. Section 6 delves into the applications of federated learning in disease diagnosis and healthcare, highlighting its associated challenges and identifying potential avenues for future research. Section 7 concludes the paper by summarizing this study's key findings, contributions, and implications.

2. Literature review

Machine learning-based Federated Learning (FL) models are emerging as transformative tools in medical diagnostics, enabling collaborative training across healthcare institutions while preserving patient

privacy. Federated Learning (FL) eliminates the necessity of sharing sensitive data across entities, thereby aligning with privacy regulations and ensuring stringent compliance with data protection standards [20, 21]. Studies reveal that FL achieves diagnostic performance comparable to centralized models, with significant improvements in tasks like next-visit predictions [22]. Personalized frameworks further enhance FL by tailoring models to local data characteristics, improving relevance for specific patient populations [20,23]. FL has demonstrated high accuracy in medical image diagnostics, achieving rates as high as 97.2% [21]. Integration with IoT devices has also facilitated efficient processing of medical images, augmenting diagnostic capabilities. Despite these advances, challenges such as data heterogeneity and the lack of standardized evaluation metrics persist [24]. Addressing these issues is critical to unlocking FL's full potential in enhancing model performance, data security, and scalability in medical diagnostics.

Dynamic client selection in federated learning (FL) is vital for addressing data heterogeneity, privacy concerns, and adversarial challenges. It ensures optimal global model performance by carefully selecting clients for each training round while preserving data privacy. Various strategies have emerged to enhance this process. FedEntOpt maximizes the entropy of the global label distribution among selected clients, reducing label skew and improving model accuracy by 6% [1]. FedDQA introduces a Data-Quality-Awareness (DQA) metric based on loss sharpness to identify high-quality data, achieving a 4% accuracy boost and faster convergence [25]. RICA incorporates data size and entropy with Centroid-Based Kernel Alignment (CKA) to exclude malicious clients, increasing accuracy by 125% in adversarial settings [2]. FANS employ a neural bandit framework, utilizing a Selection Robustness Score (SRS) to enhance resilience, boosting SRS and model accuracy under adversarial conditions [26]. These approaches highlight the advancements in FL client selection, improving robustness, efficiency, and performance in diverse scenarios. Advancements in performance, efficiency, and convergence mark the future of Federated Learning (FL) in medical diagnostics. Key challenges, such as non-IID data distributions and computational constraints, are being addressed through innovative methods. The Dynamic Data Queue-driven Federated Learning (DDFL) technique improves convergence by redistributing global subsets of data, achieving up to a 20% accuracy increase on datasets like CIFAR-100 [27]. Similarly, the Fed-RHLP approach prioritizes contributions from higher-performing clients while including lower-performing ones, enhancing accuracy and convergence in IID and non-IID settings [28]. The FedDUMAP framework leverages shared server data and dynamic updates, achieving up to 16.9× faster training and a 20.4% accuracy boost [29]. Additionally, layer-adaptive model pruning in FedDUMAP optimizes resource utilization, balancing computational efficiency and model effectiveness. These advancements pave the way for robust and scalable FL systems.

Federated learning (FL) has emerged as a pivotal approach in healthcare due to the sensitivity and distributed nature of health data [30–33]. FL addresses challenges such as inconsistent data quality, unbalanced datasets, and privacy concerns by enabling collaborative machine learning across multiple sites without requiring raw data sharing. Advanced frameworks like Federated Semi-Supervised Learning (FSSL) enhance medical image analysis, while Contribution-Aware Federated Learning (CAreFL) ensures equitable evaluation of participant contributions in innovative healthcare applications. Quantum Federated Learning (QFL) further extends FL's capabilities by leveraging quantum computing for efficient distributed training in healthcare tasks. FL proves instrumental in developing effective diagnostic models and decision support systems while ensuring compliance with regulations like GDPR and HIPAA. These advancements underscore FL's potential to revolutionize medical diagnostics and data-driven healthcare. Obtaining sufficiently large and diverse datasets for training remains a significant challenge in medical research, especially within individual institutions. Centralized patient data sharing across institutions faces privacy and data ownership hurdles. Federated learning (FL)

offers a transformative solution by enabling multi-institutional collaboration through distributed model training, allowing data to remain with its owners while only aggregating model updates. Studies demonstrate FL's effectiveness, achieving 99% model quality equivalent to centralized approaches, even when training across ten institutions [34]. Moreover, [35] conducted the most significant global FL effort to date, leveraging data from 6314 glioblastoma patients across 71 geographically diverse sites to build a robust and generalizable model for glioblastoma sub-compartment detection. Their results emphasize the importance of diverse datasets in developing clinically impactful models. Similarly, FL's utility in diagnosing hypertrophic cardiomyopathy was validated in [36], using subsets of M&M and ACDC datasets. This study confirmed FL's comparable performance to centralized systems while preserving patient privacy, underscoring its potential for privacy-preserving medical diagnostics.

Federated learning increases the accuracy of brain tumor diagnosis by facilitating collaboration without disclosing private patient information, granting access to more varied and sizable datasets [37]. Addressing the data privacy problem in healthcare settings makes it easier to create reliable systems for diagnosing fatal cancers such as brain tumors [38]. Deep learning models may be trained on decentralized data without exchanging sensitive information using federated learning methods such as Federated Averaging (FedAvg). This leads to excellent classification accuracy on independently and non-independently distributed data [39]. Furthermore, compared to conventional deep learning techniques, federated learning in brain tumor detection from MRI data has demonstrated encouraging results, delivering privacy-protected tumor classification without losing accuracy [40]. Contemporary healthcare systems gather large volumes of medical data, and data-driven machine learning (ML) has become a practical technique for creating accurate and trustworthy statistical models. The primary cause of current medical data's improper use by machine learning is that it is kept in data silos, and access to it is limited because of privacy concerns. However, without sufficient data, machine learning will not be able to reach its full potential and eventually go from research to clinical use. The authors of this article [41] highlight the challenges and aspects that must be considered while discussing how federated learning (FL) can be a solution for the future of digital health.

A consensus-based algorithm extends the Federated Learning (FL) framework to a fully decentralized environment, eliminating the need for a central server while ensuring robustness against malicious clients and low-quality data. The FedLCon algorithm enables seamless decentralization of centralized FL models like FedAvg and AdaFed, as demonstrated in COVID-19 X-ray detection tasks [42]. Additionally, a comprehensive survey highlights FL-enabling technologies, protocols, and applications, providing a detailed overview of key platforms and real-world FL use cases. This work aims to guide data scientists in designing robust, privacy-preserving FL solutions for critical sectors [43]. Implementing Federated Learning (FL) models for diabetic detection in resource-constrained healthcare settings presents several challenges due to device heterogeneity, limited computational resources, and communication inefficiencies. Many healthcare devices lack adequate processing power for local model training, while limited bandwidth can delay model updates, slowing the learning process [44, 45]. Device variability often leads to the "straggler effect", where slower devices impede overall progress, or more substantial devices dominate training, resulting in suboptimal model performance [46]. Privacy preservation remains a critical concern in under-resourced settings with weaker data security protocols [47]. Researchers are exploring solutions such as resource-adaptive algorithms and sparsity techniques to mitigate these issues [44,48], though balancing model performance with resource constraints remains a significant challenge.

3. Background study of federated learning

Federated Learning (FL) represents a decentralized paradigm for collaboratively training machine learning models across a network of distributed client devices, emphasizing preserving data privacy. Unlike traditional approaches requiring raw data centralization, FL retains the data on the client's local devices. Instead, only model updates — such as weights and gradients — are communicated to a central server. The server aggregates these updates to refine the global model iteratively.

Let N be the total number of participating clients, where each client i holds its local dataset D_i . The goal is to collaboratively train a global model w using the clients' data without centralizing it.

The overall loss function for the global model can be defined as:

$$F(w) = \sum_{i=1}^N \frac{|D_i|}{|D|} F_i(w), \quad \text{where } |D| = \sum_{i=1}^N |D_i|. \quad (1)$$

Here:

- $F(w)$ is the global loss function,
- $F_i(w)$ is the local loss function of client i evaluated on its dataset D_i ,
- $|D_i|$ is the size of the local dataset for client i ,
- $|D|$ is the total size of all clients' data.

3.0.1. Federated learning process

The FL process involves iterative communication rounds between the central server and participating clients. The steps are described as follows:

1. **Initialization:** The server initializes the global model w_0 .
2. **Local Training:** In each communication round t , the server sends the current global model w_t to the selected clients. Each client i updates the model locally by minimizing its local loss $F_i(w)$ using its dataset D_i :

$$w_i^{(t+1)} = w_t - \eta \nabla F_i(w_t), \quad (2)$$

where η is the learning rate.

3. **Model Aggregation:** After local training, the clients send their updated model weights $w_i^{(t+1)}$ to the server. The server aggregates these updates using the Federated Averaging (FedAvg) algorithm:

$$w_{t+1} = \sum_{i=1}^N \frac{|D_i|}{|D|} w_i^{(t+1)}. \quad (3)$$

4. **Global Model Update:** The aggregated model w_{t+1} is updated and sent back to the clients for the subsequent communication round.

This iterative process continues until the global model converges or meets the desired performance criteria.

3.0.2. Federated Averaging (FedAvg)

The FedAvg algorithm plays a crucial role in aggregating local models. It computes the weighted average of the locally updated models, where each client's weight is proportional to its dataset size $|D_i|$.

The mathematical formulation of FedAvg is:

$$w_{t+1} = \sum_{i=1}^N \frac{|D_i|}{|D|} w_i^{(t+1)}, \quad (4)$$

where $w_i^{(t+1)}$ is the locally trained model weight of client i in round t .

A machine learning paradigm called federated learning (FL) allows several clients (such as mobile phones, healthcare facilities, and financial institutions) to jointly train a single global model while maintaining decentralized training data. This method is beneficial when providing raw data but impractical for applications valuing privacy. The central

server manages the training procedure, compiles client-provided model updates and iteratively enhances the global model. Federated learning's primary components are (a) Clients (Participants): Hardware or other entities that store local data and carry out local model training. (b) Central Server: This central server updates the global model by combining the local models. (c) Local Training: Each client computes updates and trains the model using its local data. (d) Model Aggregation: The central server compiles the client changes to create a new global model. Algorithm 1 presented demonstrates the server-side implementation of the Federated Averaging (FedAvg) approach, a foundational technique in federated learning. FedAvg operates by iteratively aggregating local model updates from a subset of participating clients to optimize the global model. This procedure protects privacy by allowing collaborative training across dispersed data sources and maintaining client data localization. The technique coordinates critical server-side operations, such as setting up the global model, choosing clients to participate, getting local model updates, and updating the global model using weighted aggregation. The server-side implementation is pivotal in effectively managing these operations to achieve robust model performance and scalability in federated learning environments. Algorithm 2 outlines the process of client-side training within the Federated Learning (FL) framework, where individual clients independently train local models on their private data. This decentralized approach ensures that sensitive data remains on the client's device, preserving privacy while contributing to the global model's training. Each client performs several key steps during this process, including data preprocessing, local model training, and secure communication of model updates to the central server.

Algorithm 1: Federated Averaging (FedAvg).

```

1: Input: Number of clients  $K$ , number of rounds  $R$ , local epochs  $E$ , learning rate  $\eta$ 
2: Initialize: Global model parameters  $w_0$ 
3: for each round  $r = 1, \dots, R$  do
4:   Sample a subset of clients  $S_r \subseteq \{1, \dots, K\}$ 
5:   for each client  $k \in S_r$  in parallel do
6:     Download current global model  $w_r$  to client  $k$ 
7:     Update local model  $w_k^{r+1}$  using local data by solving:
        
$$w_k^{r+1} \leftarrow w_r - \eta \nabla F_k(w_r)$$

        where  $F_k(w)$  is the local loss function.
8:   end for
9:   Aggregate local models to update global model:
        
$$w_{r+1} \leftarrow \frac{1}{|S_r|} \sum_{k \in S_r} w_k^{r+1}$$

10: end for
11: Output: Global model parameters  $w_R = 0$ 

```

Explanation:

- **Initialization:** The central server initializes the global model parameters w_0 .
- **Client Sampling:** In each round r , a subset of clients S_r is randomly selected to participate in the training.
- **Local Training:** Each selected client k downloads the current global model w_r and updates it using its local data. The update is done by performing gradient descent on the local loss function $F_k(w)$.
- **Model Aggregation:** After local updates, the clients send their updated model parameters w_k^{r+1} to the central server. The server then averages these local models to form the new global model w_{r+1} .
- **Iterations:** This process is repeated for a predefined number of rounds R , resulting in the final global model w_R .

Algorithm 2: Client-Side Training.

```

1: Input: Local dataset  $D_k$ , global model parameters  $\mathbf{w}$ 
2: Output: Updated local model parameters  $\mathbf{w}_k^{(t+1)}$ 
3: Initialize local model parameters  $\mathbf{w}_k$  with  $\mathbf{w}$ 
4: for each local epoch  $e = 1, 2, \dots, E$  do
5:   Sample mini-batch  $B$  from  $D_k$ 
6:   Compute local loss:  $L_k(\mathbf{w}_k, B)$ 
7:   Compute gradient:  $\nabla L_k(\mathbf{w}_k, B)$ 
8:   Update local model:  $\mathbf{w}_k \leftarrow \mathbf{w}_k - \eta \nabla L_k(\mathbf{w}_k, B)$ 
9: end for
10: Return: Updated local model parameters  $\mathbf{w}_k^{(t+1)} = 0$ 

```

Explanation: **(a) Input and Output:** The algorithm takes the local dataset D_k and the global model parameters $\mathbf{w}$ as input and returns the updated local model parameters $\mathbf{w}_k^{(t+1)}$. **(b) Initialization:** The local model parameters $\mathbf{w}_k$ are initialized with the global model parameters $\mathbf{w}$. **(c) Local Training:** For each local epoch e , the client samples a mini-batch B from its local dataset D_k . It then computes the local loss $L_k(\mathbf{w}_k, B)$ and its gradient $\nabla L_k(\mathbf{w}_k, B)$. It updates its local model parameters $\mathbf{w}_k$ using gradient descent. **(d) Parameter Update:** The local model parameters $\mathbf{w}_k$ are updated using gradient descent with a learning rate η . **(e) Communication Overhead:** The client-side training process does not require communication with other clients or a central server. It operates solely on the client's local data. **(f) Decentralization:** Each client independently updates its local model parameters using its data, ensuring data privacy and decentralization.

A central server coordinates decentralized communication between client-side algorithms and the aggregate process in a federated learning paradigm. The global model parameters are initialized and distributed to all participating clients by the central server at the start of the workflow. Each client trains separately using its local dataset during the Client-Side Training phase. The client's training algorithm uses data and the global model from the central server to update its local model parameters. Clients send their modified model parameters — or gradients — back to the central server after local training is finished. These updates are combined by the central server using an aggregation process, which results in the updated global model parameters. The subsequent training cycle is then started by returning these modified parameters to the customers. Until the worldwide model reaches the required convergence or performance, this iterative process keeps going. To ensure data privacy and integrity, secure network protocols such as gRPC or HTTPS are typically used for communication between the clients and the central server. The central server coordinates the communication schedule, setting the deadlines for clients to submit updates, begin training, and receive new model parameters. Asynchronous communication between clients and the central server enables them to manage sporadic network access and train at their speed. Until the convergence requirements are satisfied or a certain number of rounds is reached, the client-side training and model aggregation process iterates again. While maintaining data privacy, the global model progressively improves over time based on the combined knowledge of all customers' datasets. The central server may track the federated learning process by gathering clients' convergence statistics and performance indicators. Additionally, clients can inform the central server about any problems during training or how well the global model works. To summarize, in a federated learning model, the central server and the algorithms of the clients share model updates in a decentralized manner. Iteratively dispersing the updated global model parameters to every client and coordinating the training process are the functions of the central server.

3.1. Federated learning (FL) for healthcare

The existing body of literature has yet to comprehensively explore the transformative potential of Federated Learning (FL) in healthcare.

Key areas requiring further investigation include the efficient processing of sensitive medical data, enhancing data quality, improving model precision, and effectively managing electronic health records (EHRs). Additionally, the implications of FL on critical domains such as medical imaging, the Internet of Things (IoT), and pandemic management, including the COVID-19 outbreak, remain underexamined. This paper provides a forward-looking analysis of FL applications in essential healthcare services, encompassing the management of EHRs, outbreak control (e.g., COVID-19), IoT-based health systems, and collaborative healthcare networks. The concluding section also highlights potential avenues for advancing FL in healthcare. FL facilitates collaboration among healthcare providers by enabling them to share insights while maintaining strict privacy and anonymity protocols. This capability allows the utilization of large-scale medical datasets to enhance machine learning (ML) model performance. FL is frequently applied to various tasks in the healthcare sector, including patient identification, knowledge extraction, diagnostics, predicting hospitalizations, assessing mortality risks, and analyzing intensive care unit (ICU) admission times. The development of robust ML algorithms demands access to extensive datasets, and FL provides a framework for regulated and indirect data sharing that safeguards patient privacy. FL's potential lies in enabling ML to operate on pseudonymized data, thereby addressing concerns about the accuracy of sensitive personal information. Additionally, FL environments implement distinct policies and security controls for each provider while ensuring seamless internet connectivity, further supporting secure and efficient collaborative learning in healthcare.

Federated Learning (FL) [16,49,50] has emerged as a transformative approach to addressing challenges related to data governance and privacy. By enabling collaborative training of machine learning algorithms without requiring direct data sharing, FL ensures that sensitive information remains within its originating institutions. Initially developed for applications involving mobile and edge devices [51], FL has recently gained significant traction in healthcare [52–55]. In this domain, FL facilitates the development of consensus models by aggregating insights derived from local training processes conducted independently by each participating institution. This paradigm eliminates the need to transfer patient data externally, preserving privacy and compliance with data protection regulations. Moreover, research has shown that FL-trained models can surpass the performance of models trained on centralized datasets or those developed using data from single institutions, highlighting its potential for improving outcomes in distributed and sensitive environments. Wearable technology has revolutionized healthcare by enhancing patient care, rehabilitation, and disease management. These devices generate valuable data that can aid in early detection of conditions such as dementia and diabetes. However, their widespread adoption has raised concerns regarding data personalization and privacy. To address these challenges, some other researchers have proposed a federated learning (FL) framework called FedHealth to mitigate security vulnerabilities and overcome limitations in current healthcare systems. FedHealth leverages FL and homomorphic encryption to build robust machine learning (ML) models while ensuring user privacy. As depicted in Fig. 1, the framework enables healthcare institutions — including community hospitals, research centers, disease control agencies, and training programs — to maintain and utilize extensive medical datasets. Each institution trains a shared global model locally using protected personal data, sending model updates to a centralized federated server. This server aggregates the contributions from all participants and redistributes updated parameters, enabling continued local training. This approach ensures data security while fostering collaborative medical research and patient care innovation.

Without communicating with the central location, each device or local server trains the model on its data in its environment. Consequently, the ML model is independent of the Internet connection's performance and may access sensitive data without sending it to the

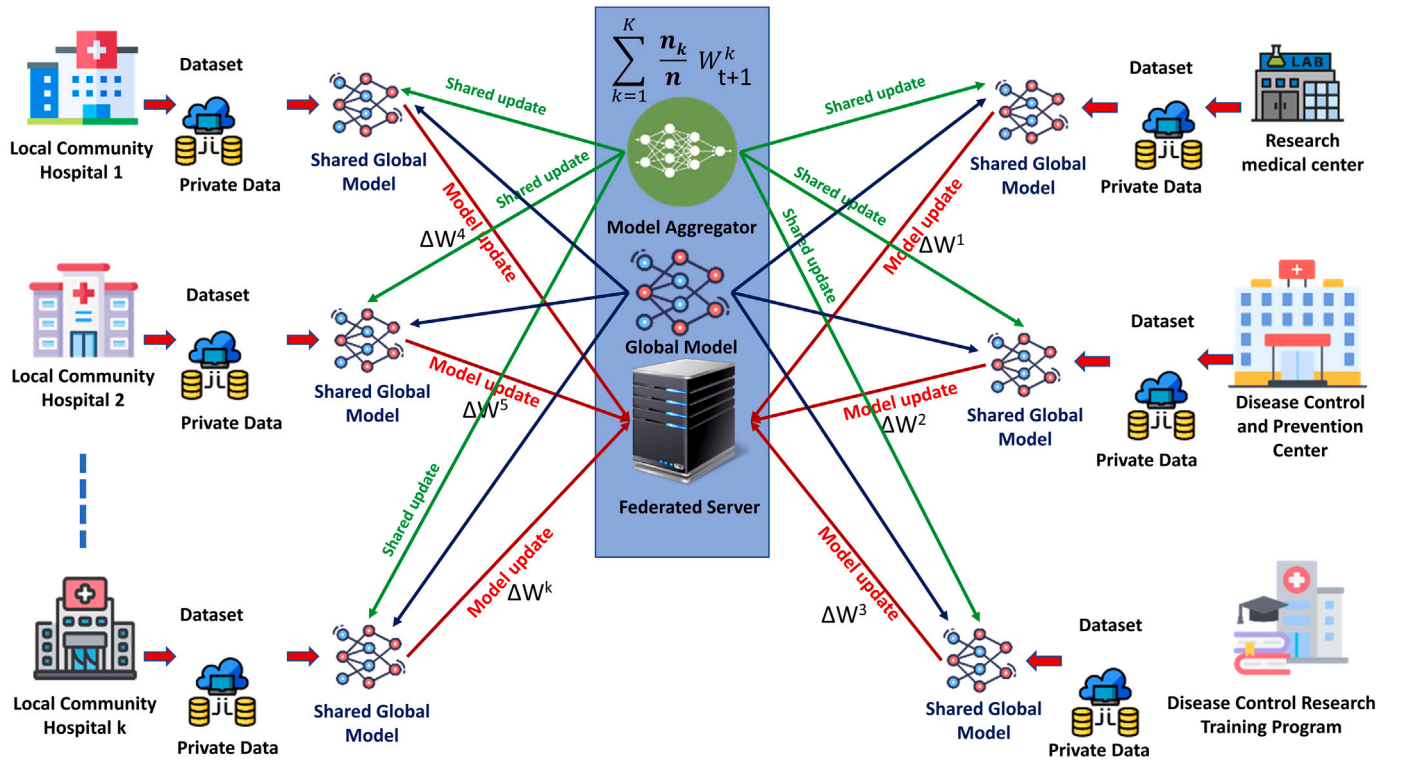


Fig. 1. The illustration demonstrates how various healthcare institutions, such as local community hospitals, medical research institutes, disease control and prevention centers, and training programs for disease control institutions, may have archives, including thousands of medical data. Each institution uses its local shared global model to train its protected personal data, and then changes are communicated to a centralized federated server. The central server and model aggregator will then combine the contributions from each participant. The updated parameters would then be provided to the participating institutes so they could continue their local training.

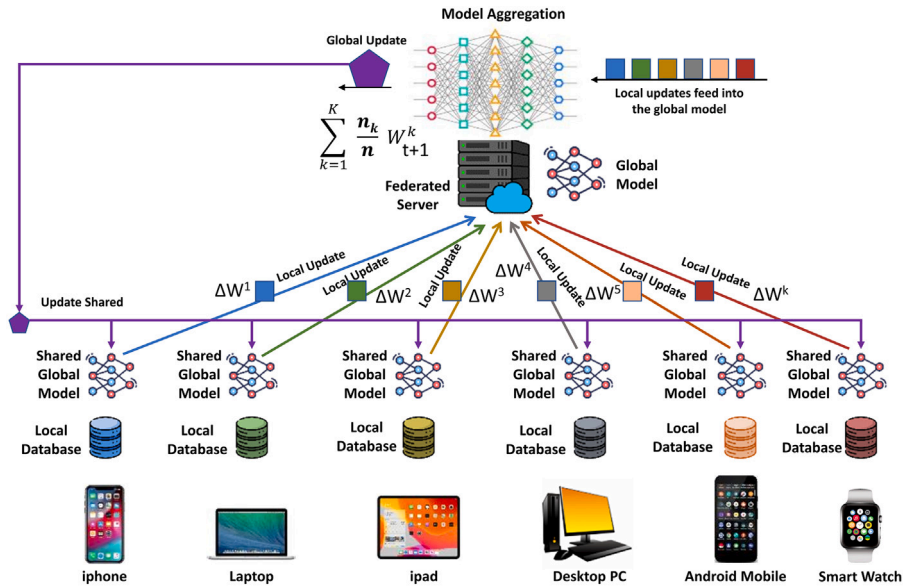


Fig. 2. In exchange for the equivalent shared updates, the device communicates the aggregation model concerning its local shared global model parameters. Then, it enhances its local characteristics by combining its received inputs. Up until convergence, this process is repeated.

cloud. Fig. 2 illustrates the exchange process in a federated learning model among various local devices, including tablets, phones, PCs, smartwatches, IoT devices, and wearables. Each device communicates shared updates and aggregates the global model parameters with its local model. It then enhances its local characteristics by integrating the received inputs. This iterative process continues until convergence. The teaching of AI models could be revolutionized by federated learning, which would also benefit the broader healthcare system. More

broad hospital networks would benefit from improved collaboration if they could access secure, cross-institutional data. High-level artificial intelligence algorithms would be available to smaller hospitals in rural and community settings. AI may reach the stage of therapy if it is made easier to include vast volumes of diverse data from multiple organizations in the model-building process while maintaining local control over the clinical data. More trustworthy AI algorithms would be available to clinicians based on data from rare cases they would not

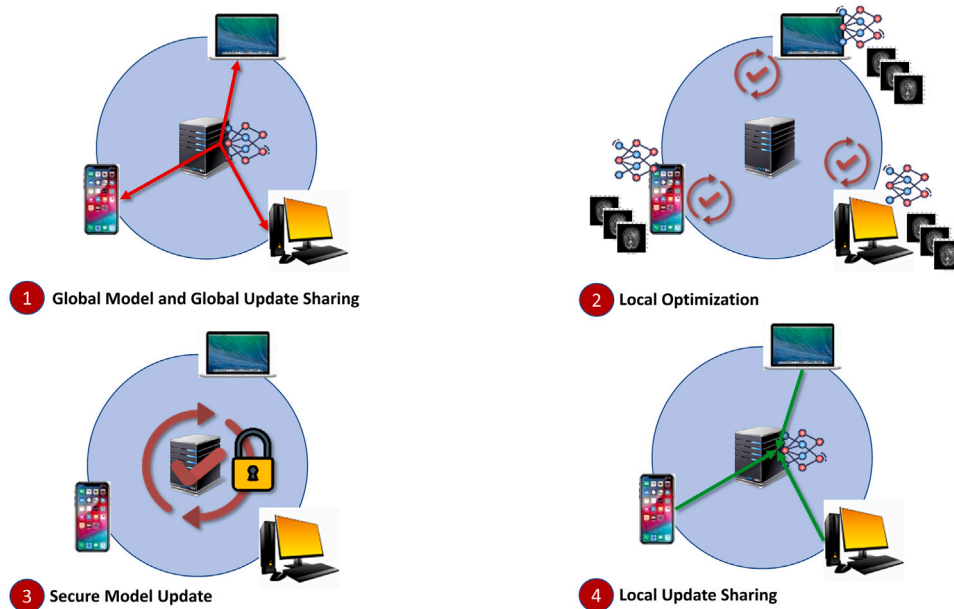


Fig. 3. Federated Learning updating steps. The first step is the global model and update sharing, the second step is local optimization, the third is a secure model update, and the fourth is local update sharing.

have encountered locally or from data covering more important patient demographics for a particular therapeutic area. When they disagree with the results, they may also contribute back to the training of these algorithms. Thanks to a safe method for learning from various algorithms, healthcare firms might introduce ground-breaking inventions to the market more quickly. The restricted number of free datasets would be replaced with a vast array of real-world data, allowing research institutes to focus their efforts on pressing clinical needs.

Although these data sources are typically kept separate, individual healthcare institutes may have archives with hundreds of thousands of records and photographs. This is primarily because health information is confidential and cannot be utilized without patient permission and ethical authorization. Federated learning decentralizes deep learning by eliminating the requirement to pool data in a single location. Instead, the model is trained across several iterations at various locations. A centralized server would manage the global deep neural network in a client-server federated architecture, and each participating hospital would receive a copy to train on its dataset. The participants would maintain their dataset in their private infrastructure and submit their updated model version to the centralized server once the model had been trained locally for a few iterations. The central server would then combine the contributions from each participant. The participating institutes would then get the revised parameters for their local training. Since the model is not dependent on any specific data, if one of the hospitals decided it wanted to quit the training team, this would not stop the model's training. The project is also open to additional hospitals at any point deciding to participate.

Step 1: The central or global model is distributed to all edge nodes by storing local network copies. The global updates are likewise distributed to all local devices following the global model aggregations. Step 2: The local networks are updated by learning from local data samples. The model's neural network weights include the "learning" information. Reversing this process would require taking these weights and decoding them to obtain the original data, which is problematic. Then, the encrypted learned model weights are used. To transmit to the central server. Step 3: All local modifications are encrypted and sent to the central server. Step 4: The central server receives the weight updates from all local models and combines them without decoding the weights to rebuild the original data (which is very challenging) or remove the encryption (based on homomorphic encryption encryptions).

During the aggregation, the central model considers how many samples each local used to update its model. More data more significantly impacts the whole model. To return to step 1, share the updated network with every local device. Fig. 3 illustrates the steps of Federated Learning: (1) Global model and update sharing, (2) Local optimization, (3) Secure model updates, and (4) Local update sharing.

4. Materials and methods

4.1. Client selection in federated learning

A novel approach to cooperatively training machine learning models while protecting data privacy is federated learning (FL). In contrast to conventional centralized methods, FL allows a heterogeneous network of clients, each with its local datasets—to work together to construct a global model without exchanging raw data. In addition to improving privacy and data security, this decentralized structure allows devices with limited processing and storage capabilities to participate in the training process. A critical aspect of FL is client selection, which determines the subset of clients participating in each training iteration. However, efficient client selection is particularly challenging due to the inherent heterogeneity of clients. Variations in data distribution, computational capabilities, and network conditions create complexities in identifying an optimal selection strategy. These challenges underscore the importance of developing robust client selection mechanisms to enhance model performance, accelerate convergence, and ensure system efficiency. Overcoming these obstacles is essential for FL to realize its full potential as a scalable, privacy-preserving solution for collaborative machine learning.

Because it establishes the subset of participants in each training iteration, client selection is essential to the Federated Learning (FL) process. Finding a team that can produce high-quality updates to the global model is the main objective of client selection, taking into account variables like communication overhead, computing efficiency, resource availability, and data quality. The meticulous client selection process dramatically impacts the FL system's overall efficacy, convergence speed, and performance. The central server selects a subset of clients from the accessible pool using predetermined techniques or criteria in this phase, which takes place before model startup and local training. These selection procedures can take many forms, from

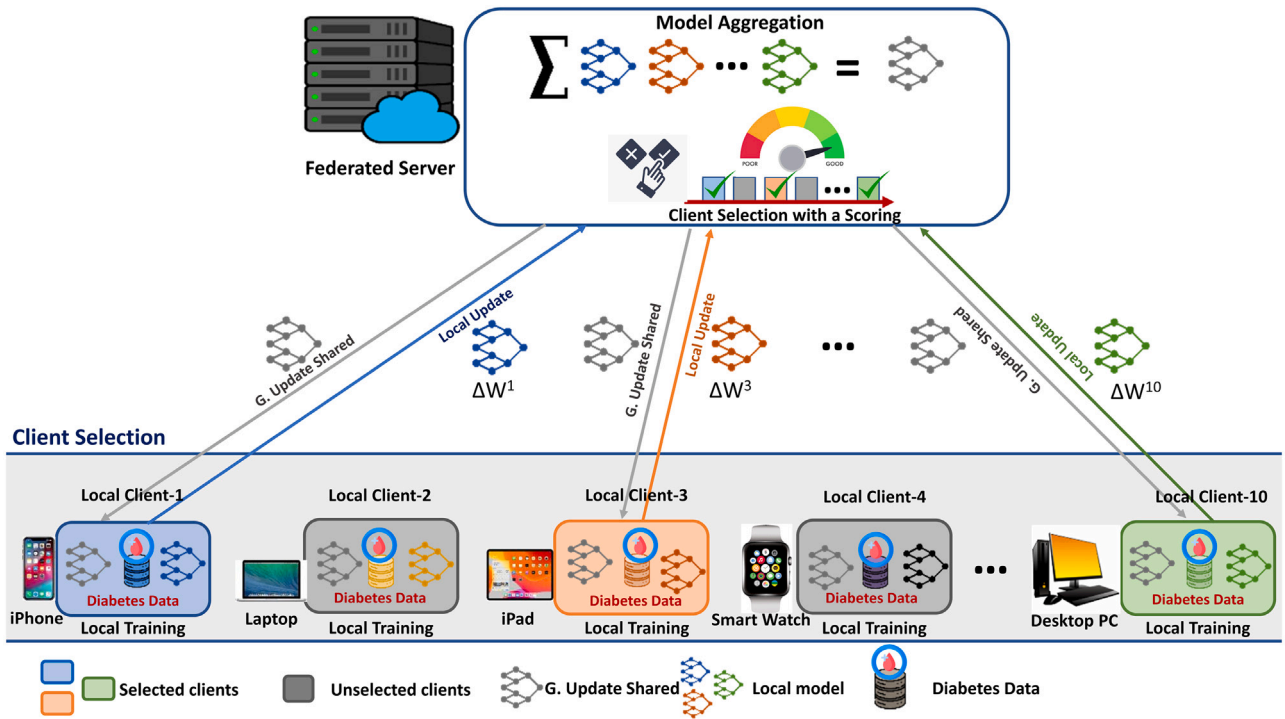


Fig. 4. Dynamic client selection in federated learning.

essential random sampling to more sophisticated techniques considering important variables like client data distribution, computational resource limitations, network latency, and past performance indicators. For instance, performance-based methods can provide preference to clients that offer updates with greater precision. In contrast, resource-aware strategies might prefer clients with better computing capabilities and reduced network latency.

In federated learning (FL), the efficient client selection process dramatically influences the system's effectiveness and performance. Selecting clients with high-quality, varied datasets and adequate processing power helps hasten the global model's convergence and improves its capacity to generalize in various situations. On the other hand, poor client selection can lead to increased communication delays, biased model updates, and delayed convergence, all of which reduce the FL framework's overall efficacy. Poorly chosen clients might endanger the model's privacy and security in certain situations. Recent studies have concentrated on creating sophisticated client selection techniques that consider the complications brought on by client and system heterogeneity to lessen these difficulties. By combining several elements like data variety, resource availability, and equitable contributions, these methods seek to maximize client selection to enhance the robustness and performance of federated learning systems.

Fig. 4 depicts the process of dynamic client selection within a federated learning (FL) framework. This approach utilizes a scoring system to assess and rank clients according to key performance metrics, such as accuracy, execution time, and loss. In each communication round, scores are calculated for all participating clients, and metrics are normalized to ensure equity, leading to the dynamic selection of the most suitable clients for training. By incorporating this scoring mechanism, the FL method optimizes model performance, accelerates convergence speed, and ensures system efficiency. The dynamic selection process accommodates client heterogeneity, effectively managing variations in data distribution, computational power, and network conditions. The figure highlights the essential role of adaptive client selection in achieving a robust and efficient federated learning system.

A key component of the Federated Learning (FL) system is client selection, which chooses which subset of clients will participate in

each training session. Client selection aims to find customers who can deliver high-quality updates to the global model while considering communication effectiveness, resource availability, and data quality. Choosing clients wisely is crucial to maximizing FL systems' effectiveness and performance. Involving clients with high-quality data and adequate computing power speeds up the global model's convergence and improves its capacity for generalization. On the other hand, improper client selection can slow down the learning process, introduce biases, and increase the risks associated with security and privacy. As a core component of FL, client selection directly influences the system's performance, efficiency, and fairness. Several obstacles must be overcome to identify the optimal client set for every round, including statistical heterogeneity, system capabilities, communication limitations, bias, fairness, and resilience to security attacks. These issues must be resolved to prioritize clients and enhance FL systems' overall performance. Ongoing research efforts are focused on developing strategies to balance resource limitations, improve fairness, and mitigate data heterogeneity, aiming to establish more robust, efficient, and secure federated learning systems.

4.1.1. Scoring criteria

The scoring criteria for dynamic client selection in federated learning focus on balancing performance (accuracy and loss), efficiency (execution time and resource availability), fairness, and adaptability. By incorporating normalized metrics, dynamic weight adjustment, and historical contributions, the scoring system ensures optimal client participation while addressing challenges such as heterogeneity, resource constraints, and communication overhead. This approach significantly improves the federated learning model's overall performance, convergence speed, and robustness. Some critical scoring criteria for dynamic client selection in federated learning are outlined below:

- **Model Improvement:** (accuracy or loss reduction, proper selection of learning algorithm) The performance improvement observed when incorporating a client's updates. Model performance improvement heavily depends on the appropriate choice of learning algorithms. To achieve optimal results, the selection of learning algorithms within FL paradigms can leverage our proposed Federated Adaptive Learning approaches.

- **Execution Time:** Execution time is when the client trains its model and returns results to the server locally. Efficient clients with faster execution times improve the overall speed of the training process. Since execution times vary significantly across clients, they must be normalized to avoid outlier dominance.
- **Data Quality:** Quality, quantity, data heterogeneity, imbalanced data, and diversity of the client's local dataset.
- **Convergence Speed:** The rate at which the client's model reaches a local optimum during training.
- **Resource Availability:** The computational and communication resources (e.g., CPU, memory, and network bandwidth) available to each client. Clients with better resource availability can complete their tasks faster and without interruption, improving system efficiency. Clients with limited resources may be deprioritized to avoid delays, though their participation should not be ignored entirely for fairness. The resources we must consider include computational power, CPU, memory, network bandwidth, latency, communication efficiency, and client energy constraints.
- **Contribution Consistency:** The historical performance and contribution of each client in previous rounds (accuracy improvement, loss reduction, or execution time). Clients that consistently perform well can be prioritized, ensuring more reliable updates for the global model. Dynamic scoring can increase weight factors (e.g., α for accuracy) for consistently improving clients.
- **Fairness and Participation Balance:** Ensuring that all clients have opportunities to participate in training, even if they have relatively lower scores. Bias towards high-performing clients can lead to model bias and neglect valuable data from underrepresented clients. Introduce participation quotas or lower-bound thresholds to include a mix of high and low-performing clients.
- **Communication Cost:** The cost of transmitting model updates between the central server and the client. Clients with higher communication latency or expenses can be deprioritized to save system bandwidth and improve efficiency—balance communication cost with performance to avoid excluding resource-limited but valuable clients.
- **Dynamic Weight Adjustment:** Adjusting the scoring system's weight factors (α , β , γ) dynamically based on current client behavior and performance trends. Ensures the scoring system remains adaptive to real-time conditions and client performance fluctuations.
- **Client Dropout Probability:** The likelihood of a client failing to complete training or dropping out due to unstable connections, limited resources, or power constraints. Predicting dropout probability helps prioritize reliable clients, reducing failed training rounds. Implement mechanisms to mitigate dropout risks, such as client redundancy or backup selections.
- **Impact on Global Model Performance:** Evaluating the impact of each client's updates on the performance of the global model in previous rounds. Clients whose updates positively influence the global model's accuracy or convergence should be prioritized. Introduce feedback mechanisms to adjust client scores based on their contributions to global model improvement.
- **Previous Record:** Previous scores and participation of the client.

4.2. Dataset

The data set “diabetes binary health indicators BRFSS2015” consists of 253,680 responses collected from the Centers for Disease Control and Prevention (CDC) through the 2015 behavioral risk factor surveillance system (BRFSS) [56]. The target variable in this dataset, diabetes binary, is categorized into two groups: 0 indicates that the individual has neither diabetes nor prediabetes. In contrast, a value of 1 indicates the presence of either condition. The dataset is imbalanced, with a higher prevalence of individuals categorized as 0. It includes 21 feature variables that provide insights into various health indicators

and lifestyle factors. The BRFSS, conducted annually by the CDC since 1984, is a nationwide telephone survey with participation from over 400,000 adults in the United States each year. The survey collects data on various health-related behaviors, chronic health conditions, and the use of preventive healthcare services. Although diabetes currently has no definitive cure, it is a condition that can be managed and, in some cases, prevented through lifestyle interventions such as weight management, healthy eating, regular physical activity, and access to appropriate medical care.

Predictive models for assessing diabetes risk are crucial for the general population and public health professionals, as early detection can lead to timely lifestyle modifications and more effective management of the condition. Such models provide valuable insights that can facilitate preventive measures and improve treatment outcomes. A comprehensive overview of the dataset's attributes and corresponding descriptions is provided in Table 1, offering a detailed understanding of the data utilized in the analysis.

4.3. Proposed methods

In this work, we employed the Random Forest machine learning algorithm as the global model for diabetic detection, integrated into a dynamic federated learning framework. The framework implements a dynamic client selection strategy with a scoring system to optimize performance and enhance system efficiency. The scoring system enables intelligent client selection during each communication round, ensuring that the most suitable clients contribute to the global model updates. The training and evaluation of the model were performed using the diabetes binary health indicators BRFSS2015 dataset [56], which consists of 253,680 survey responses collected by the CDC. This dataset provides a robust foundation for training a machine learning model to predict diabetes-related outcomes.

The data was distributed among ten local devices, simulating federated learning clients. The data set was first divided into training and testing sets, and the training data was randomly partitioned between the ten local devices. Each device independently trained the model on its local subset of data, preserving privacy by not sharing raw data with the central server. The global model aggregates the updates of the local models using the federated learning paradigm.

To further enhance the learning process, a Client Scoring and Selection Algorithm was employed to dynamically select clients in each communication round based on their performance metrics: accuracy, execution time, and loss. The scoring formula considers these metrics with weights α , β , and γ assigned to accuracy, execution time, and loss, respectively. Execution time and loss values are normalized to ensure fair scoring among clients, accounting for variations in computational resources and data complexity. The algorithm ensures that participation is prioritized for clients with higher accuracy, lower execution time, and reduced loss, thereby improving the overall model performance and training efficiency. The weights α , β , and γ are dynamically adjusted based on observed trends in client performance, enabling the system to adapt to changes and continuously optimize the global learning process. This dynamic selection approach, combined with the robust capabilities of the Random Forest algorithm, ensures effective and efficient training of the international model for diabetic detection.

4.3.1. The proposed approach for dynamic client selection using a scoring system

We adopted a systematic approach to optimize the selection of participating client devices in each communication round to implement dynamic client selection with a scoring system in a federated learning framework. This approach evaluates each client based on key performance metrics: accuracy, execution time, and loss. Incorporating these metrics into a comprehensive scoring formula ensures that only the most effective and efficient clients are selected to contribute to the global model update during each round. This method prioritizes

Table 1
Dataset properties and description.

Properties	Description
Dataset name	The dataset utilized in this study is titled “Diabetes Binary Health Indicators from the Behavioral Risk Factor Surveillance System (BRFSS) 2015”.
Data types	The diabetic dataset comprises information from over 400,000 individuals in the United States, detailing their engagement with preventive healthcare services, participation in risky behaviors, and the prevalence of chronic health conditions.
Dataset source	The Behavioral Risk Factor Surveillance System (BRFSS) is an annual telephone-based survey conducted by the Centers for Disease Control and Prevention (CDC) to collect data on chronic diseases, health behaviors, and preventive practices. This information is used to shape public health policies and programs. A publicly available version of the dataset on Kaggle serves as a valuable resource for data-driven research and health analytics.
Context	Diabetes has become a global health concern, affecting individuals of all age groups at an alarming rate. If left unmanaged, this chronic condition can lead to severe complications, contributing to the dysfunction and failure of vital organs such as the liver, kidneys, heart, and gastrointestinal system. These complications significantly increase the mortality risk, highlighting the critical need for effective management and prevention strategies.
Total rows and columns	The dataset comprises 253,681 rows and 22 columns, representing a substantial volume of records and features for analysis.
Attributes	Diabetes binary, HighBP, HighChol, CholCheck, BMI, Smoker, Stroke, HeartDisease or Attack, Physical Activity, Fruits and Vegetables, Heavy Alcohol Consumption, Any Healthcare, No Doctor's Visit Cost, General Health, Mental Health, Physical Health, DiffWalk, Sex, Age, Education, and Income.
Target level	The target variable, Diabetes_binary, is a binary classification where 0 indicates the absence of diabetes, and 1 represents prediabetes or diabetes. A value of 1 signifies elevated blood glucose levels associated with these conditions.

clients demonstrating high model accuracy, minimal training time, and low loss, enabling the federated learning process to achieve improved performance and efficiency over time. The proposed approach for dynamic client selection utilizing a scoring system is outlined below:

(a) Score Calculation: To improve the client selection score based on accuracy, execution time, and loss, we use the formula:

$$\text{Score}_i = \alpha \cdot \text{accuracy}_i - \beta \cdot \text{Normalized_Execution_Time}_i - \gamma \cdot \text{Normalized_Loss}_i \quad (5)$$

where:

- α (Alpha) is a weight factor for accuracy,
- β (Beta) is a weight factor for execution time,
- γ (Gamma) is a weight factor for loss,
- **accuracy** is the client's model accuracy,
- **execution_time** is the time taken for the client to train its model,
- **loss** is the model loss.

(b) Normalize or Scale the Features:

Execution Time: Execution times may vary drastically depending on the client's computational resources. Normalizing or scaling execution time to avoid outliers is critical for fair scoring.

$$\text{normalized_execution_time} = \frac{\text{times.max() - times}}{\text{times.max() - times.min()}} \quad (6)$$

Loss: Like accuracy, loss values can vary significantly between clients. Normalizing loss or setting a threshold is essential to avoid highly high loss values that negatively affect the score.

$$\text{normalized_loss} = \frac{\text{losses.max() - losses}}{\text{losses.max() - losses.min()}} \quad (7)$$

(c) Dynamic Weight Adjustment for α , β , and γ : Adjusting α , β , and γ dynamically based on the performance of previous rounds can help better prioritize the clients. For instance:

- If the accuracy of a client improves over time, increase α to give more weight to accuracy.

- If the execution time for a client becomes a bottleneck (e.g., it takes longer to converge), increase β to penalize the long execution time more.
- If the loss decreases steadily for a client, reduce γ to minimize the penalty for loss.

```
if client_accuracy > previous_accuracy:
# Increase alpha if accuracy improves
if client_execution_time > threshold_time:
# Increase beta if execution time exceeds
a threshold if client_loss > previous_loss:
# Increase gamma if loss increases
```

(d) Implement Client Evaluation and Selection Mechanism: Sort clients based on their score: After calculating the scores for all clients using the formula, we can sort them and select the top clients for the next training round.

(e) Add Execution Time Penalty for Long Rounds: If execution time increases significantly, consider adding a penalty to the execution time factor in the score calculation:

```
if client.execution_time > max_execution
_time_threshold: score -= penalty_factor*
(client.execution_time - max_execution_
time_threshold)
```

(f) Monitor and Adjust Over Time: Continuously monitor the impact of the score adjustment strategy. Over time, fine-tune parameters (α , β , γ) based on real-world performance to balance accuracy, execution time, and loss better.

By applying these strategies, you can improve client selection by balancing the contribution of accuracy, execution time, and loss to ensure efficient training in federated learning systems.

(g) Dynamic Weight Adjustment Formulation: To enhance adaptability and fairness in client selection, the scoring weights α , β , and γ

are dynamically adjusted over communication rounds based on client performance trends. The weights are updated using moving average trends in accuracy, execution time, and log loss over the past few rounds.

Let δ be a small learning rate (e.g., $\delta = 0.01$). For each round r , define the 3-round moving average of accuracy, execution time, and loss as:

$$\overline{\text{Acc}}^{(r)} = \frac{1}{3} \sum_{j=r-2}^r \text{Acc}^{(j)} \quad (8)$$

$$\overline{\text{Time}}^{(r)} = \frac{1}{3} \sum_{j=r-2}^r \text{Time}^{(j)} \quad (9)$$

$$\overline{\text{Loss}}^{(r)} = \frac{1}{3} \sum_{j=r-2}^r \text{Loss}^{(j)} \quad (10)$$

The scoring weights are updated based on whether the recent trend indicates improvement or degradation:

$$\alpha^{(r+1)} = \begin{cases} \alpha^{(r)} + \delta, & \text{if } \overline{\text{Acc}}^{(r)} > \overline{\text{Acc}}^{(r-1)} \\ \alpha^{(r)} - \delta, & \text{otherwise} \end{cases} \quad (11)$$

$$\beta^{(r+1)} = \begin{cases} \beta^{(r)} + \delta, & \text{if } \overline{\text{Time}}^{(r)} > \overline{\text{Time}}^{(r-1)} \\ \beta^{(r)}, & \text{otherwise} \end{cases} \quad (12)$$

$$\gamma^{(r+1)} = \begin{cases} \gamma^{(r)} + \delta, & \text{if } \overline{\text{Loss}}^{(r)} > \overline{\text{Loss}}^{(r-1)} \\ \gamma^{(r)}, & \text{otherwise} \end{cases} \quad (13)$$

This approach ensures that:

- α increases when model accuracy improves, emphasizing performance.
- β increases when execution time increases, penalizing inefficiency.
- γ increases when model loss increases, penalizing poor convergence.

The weights remain stable unless triggered by significant performance trends, providing dynamic yet controlled adaptability in federated client selection.

The Algorithm for dynamically client scoring and selection in Federated Learning aims to optimize the selection of participating client devices in each communication round by considering their performance metrics: accuracy, execution time, and loss. The algorithm begins by initializing the weight parameters α , β , and γ , which represent the importance of accuracy, execution time, and loss, respectively. Before proceeding with communication rounds, the execution time and loss values of all clients are normalized to ensure fair comparison and avoid bias caused by variations in computational resources or outlier values. Normalization uses a min-max scaling formula to standardize the metrics across all clients. Using the formula, the algorithm calculates a score for each client in each communication round. The scores are stored, and clients are sorted in descending order based on their calculated scores. The top k clients with the highest scores are selected to participate in the next round. This selection process ensures that high-performing clients contribute more significantly to the global model updates. Additionally, the algorithm dynamically adjusts the weights α , β , and γ after each communication round to reflect changes in client performance trends. For instance, α is increased if a client's accuracy improves over time, β is raised if execution times exceed a predefined threshold, and γ is adjusted based on the loss trends observed across rounds. This dynamic adjustment enables the algorithm to adaptively balance accuracy, computational efficiency, and model loss during client selection. The process iterates through all communication rounds, continuously updating the global model with contributions from the selected clients. The final global model is produced at the end of all rounds, representing the aggregate learning across all assigned clients. This approach enhances the efficiency and performance of the federated

learning framework by prioritizing clients that optimize the balance between accuracy, execution time, and loss. Algorithm 3 illustrates our proposed approach for dynamic client selection with a scoring system in the context of Federated Learning. This method prioritizes clients based on a scoring mechanism to evaluate multiple criteria. The proposed system aims to optimize overall model performance, enhance convergence speed, and ensure equitable participation across heterogeneous clients by dynamically selecting the most suitable subset of clients for each training round.

Algorithm 3: Dynamic Client Selection with a Scoring System.

Require: Dataset (X, y) ; number of clients N ; communication rounds R ; scoring weights α, β, γ .

- 1: **Initialize:** Randomly split data among N clients:
 $(X_1, y_1), (X_2, y_2), \dots, (X_N, y_N)$
- 2: Initialize client scores: $S_i \leftarrow 0, \forall i \in [1, N]$
- 3: Initialize global model $\mathcal{G}$ (e.g., Random Forest Classifier)
- 4: **for** each communication round $r = 1$ to R **do**
- 5: **Dynamic Client Selection:**
- 6: Select top $k = \frac{N}{2}$ clients based on the highest scores
- 7: $C_r \leftarrow$ Set of selected clients
- 8: **for** each client $i \in C_r$ **do**
- 9: Split client data (X_i, y_i) into training and testing:
 $(X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}})$
- 10: **Train Local Model:**
- 11: $\mathcal{L}_i \leftarrow$ Train a Random Forest Classifier on $(X_{\text{train}}, y_{\text{train}})$
- 12: Measure **Execution Time:** time_i
- 13: Evaluate local model:
- 14: $\text{accuracy}_i \leftarrow$ Accuracy Score
- 15: $\text{loss}_i \leftarrow$ Log Loss
- 16: **end for**
- 17: **Normalize Execution Time and Loss:**
- 18: Normalize time_i and loss_i using defined equations
- 19: **Update Client Scores:**
- 20: **for** each client $i \in C_r$ **do**
- 21: $S_i \leftarrow \alpha \cdot \text{accuracy}_i - \beta \cdot \text{Normalized_Execution_Time}_i - \gamma \cdot \text{Normalized_Loss}_i$
- 22: **end for**
- 23: **Global Model Update:**
- 24: Aggregate local models $\mathcal{L}_i$ using weighted averaging
- 25: Update global model $\mathcal{G}$
- 26: Evaluate global model $\mathcal{G}$ on global test data $(X_{\text{global}}, y_{\text{global}})$
- 27: **end for**
- 28: **Output:** Trained global model $\mathcal{G} = 0$

Algorithm 4 illustrates Dynamic Client Selection with Adaptive Scoring Weights, an intelligent client selection strategy in Federated Learning (FL) that prioritizes clients based on performance trends over multiple communication rounds. Instead of selecting clients randomly, this method dynamically adjusts selection probabilities using a **scoring function** that considers key performance metrics such as *accuracy*, *execution time*, and *loss*. The scoring weights (α, β, γ) are **adaptively updated** using moving averages, ensuring that high-performing clients are selected while maintaining fairness and robustness. This approach enhances **model convergence**, **resource efficiency**, and **resilience** to data quality variations, making it a more effective and scalable solution in Federated Learning.

We calculated the standard classification and regression metrics used to evaluate the performance of our MLP model. These metrics help understand the model's effectiveness and accuracy in different contexts. Evaluation Metrics for Multilayer Perceptron (MLP) are Accuracy, Precision, Recall (Sensitivity), F1 Score, ROC-AUC (Receiver Operating Characteristic - Area Under Curve), and Confusion Matrix.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (14)$$

Algorithm 4: Dynamic Client Selection with Adaptive Scoring Weights

Require: Dataset (X, y) ; number of clients N ; rounds R ; initial weights α, β, γ

- 1: Initialize: Split data among N clients: $(X_1, y_1), \dots, (X_N, y_N)$
- 2: Initialize client scores: $S_i \leftarrow 0, \forall i \in [1, N]$
- 3: Initialize global model $\mathcal{G}$
- 4: Initialize lists to track trends: $\text{AccHist}_i, \text{LossHist}_i, \text{TimeHist}_i$
- 5: **for** each round $r = 1$ to R **do**
- 6: Select top $k = \frac{N}{2}$ clients based on highest scores $\rightarrow C_r$
- 7: **for** each client $i \in C_r$ **do**
- 8: Train local model $\mathcal{L}_i$ on local train split
- 9: Record metrics: accuracy $_i$, loss $_i$, exec_time $_i$
- 10: Update history: Append to $\text{AccHist}_i, \text{LossHist}_i, \text{TimeHist}_i$
- 11: **end for**
- 12: Normalize loss and time over selected clients
- 13: **for** each client $i \in C_r$ **do**
- 14: $S_i \leftarrow \alpha \cdot \text{accuracy}_i - \beta \cdot \text{norm_time}_i - \gamma \cdot \text{norm_loss}_i$
- 15: **end for**
- 16: **Adjust Weights Based on Trends:**
- 17: **if** Average accuracy in last 3 rounds increased **then**
- 18: Increase $\alpha \leftarrow \alpha + \delta$
- 19: **else**
- 20: Decrease $\alpha \leftarrow \alpha - \delta$
- 21: **end if**
- 22: **if** Average loss in last 3 rounds increased **then**
- 23: Increase $\gamma \leftarrow \gamma + \delta$
- 24: **end if**
- 25: **if** Average exec_time in last 3 rounds increased **then**
- 26: Increase $\beta \leftarrow \beta + \delta$
- 27: **end if**
- 28: Aggregate $\mathcal{L}_i$ into global model $\mathcal{G}$
- 29: Evaluate $\mathcal{G}$ on $(X_{\text{global}}, y_{\text{global}})$
- 30: **end for**
- 31: **Output:** Final global model $\mathcal{G} = 0$

4.4. Hyperparameter selection and experimental justification

We independently varied the number of participating clients and the initial scoring weights (α, β, γ) to examine their individual contributions to the model's performance. Our sensitivity analysis revealed that while increasing the number of clients generally enhances model robustness, the dynamic evolution of α, β, γ across the 200 communication rounds is critical in balancing accuracy, execution time, and loss. For instance, adjusting α , which prioritizes accuracy—demonstrated a direct correlation with improved global accuracy, as reflected in the corresponding performance metrics and the ROC curve. Similarly, modifications to β and γ were observed to impact convergence behavior and loss trends without significantly detracting from the overall performance. Overall, these findings confirm that our federated learning framework not only adapts to varying hyperparameter settings but also maintains strong generalizability and robustness under different experimental conditions. The details of the hyperparameter selection technique are shown in Table 2.

5. Results and discussions

A thorough summary of the standard hardware and software requirements used in simulation settings is provided in Table 3. It highlights key components such as the type of processors (CPUs), memory capacity (RAM), storage availability, and hardware accelerators like GPUs and TPUs, which are essential for handling computationally intensive tasks. Additionally, the table outlines the software stack,

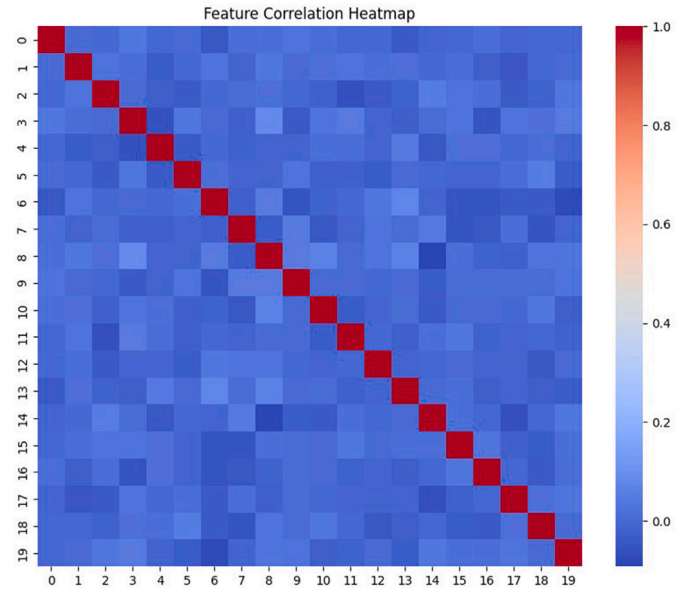


Fig. 5. The feature correlation heatmap.

including operating systems, programming languages, and preinstalled libraries, which collectively support efficient simulation execution. The specifications are categorized to showcase variations across different configurations, ensuring flexibility and adaptability for diverse simulation workloads.

We utilized the Diabetes Binary Health Indicators BRFSS2015 dataset [56], collected initially through a survey conducted by the Centers for Disease Control and Prevention (CDC) to address health-related issues. This dataset is publicly available on Kaggle and comprises 253,681 records with 22 distinct features, making it suitable for analyzing various health indicators. We partitioned the dataset into ten local client devices to simulate a federated learning environment. The data was randomly and evenly distributed across these client devices to ensure a balanced yet realistic simulation of decentralized data storage. Each local client device performed local training independently using its respective subset of the raw data, stored locally without sharing data between devices. We employed the Federated Averaging (FedAvg) algorithm for the global model aggregation, which combines the locally trained models to produce a centralized, federated outcome.

The heatmap illustrates the correlation between features, with the diagonal elements exhibiting a correlation value of 1, indicating perfect self-correlation in Fig. 5. The predominantly blue off-diagonal elements suggest weak or negligible correlations among most features, signifying minimal multicollinearity within the dataset. The absence of strong off-diagonal clusters further indicates that the features are primarily independent, contributing to a well-structured dataset with minimal redundancy. Moreover, the confusion matrix and the Receiver Operating Characteristics (ROC) curve are represented in Fig. 6.

The global accuracy and loss evolution of a federated learning-based machine learning model for diabetes detection over 200 communication cycles are shown in Fig. 7. This framework employs a dynamic client selection mechanism, leveraging a scoring system that integrates metrics such as accuracy, execution time, and loss for each client. The global accuracy curve demonstrates a consistent upward trend, reflecting the model's improving predictive performance as communication rounds progress. Meanwhile, the global loss curve steadily declines, indicating enhanced model optimization and reduced prediction error. The dynamic client selection strategy ensures that high-performing clients contribute more effectively, thus accelerating model convergence and improving overall efficiency. Through decentralized training, this visualization demonstrates how flexible the federated learning

Table 2
Hyperparameter settings for the federated learning experiment.

Hyperparameter	Description	Value
α	Initial Weight for accuracy (positive impact)	0.5
β	Initial Weight for execution time (negative impact)	0.3
γ	Initial Weight for loss (negative impact)	0.2
num_clients	Number of participating clients	10
communication_rounds	Number of federated learning rounds	200
client_selection_rate	Fraction of clients selected each round	50% (Top 5)
n_estimators	Trees in global Random Forest	≤ 100
test_size (local)	Local client test data proportion	0.2 (20%)
test_size (global)	Global evaluation dataset size	50,736 samples (20%)
random_state	Random seed for reproducibility	42
normalization_epsilon	Epsilon value to prevent division by zero	1×10^{-5}
num_features	Number of features per data sample	20
total_samples	Total number of data samples	253,680

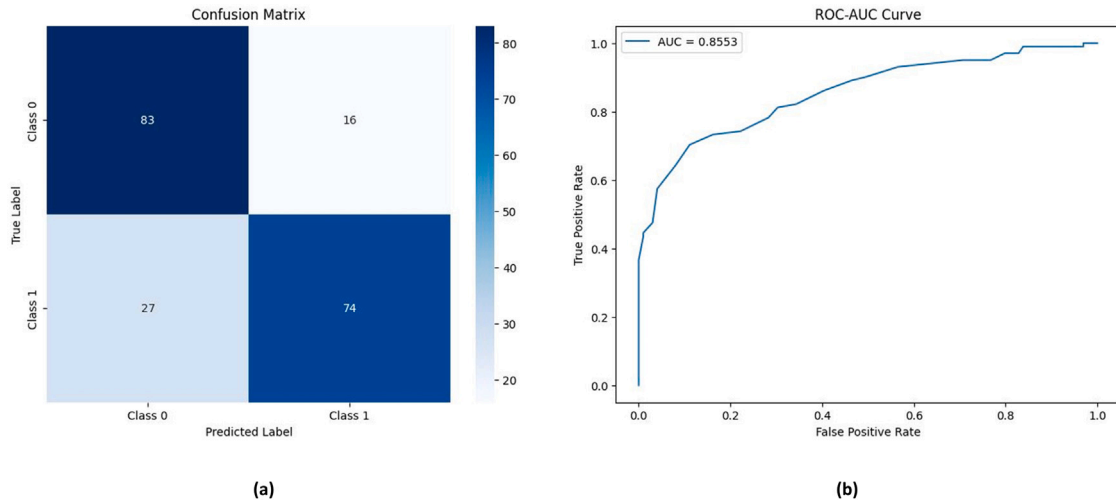


Fig. 6. Performance metrics of Random Forest (RF) in the proposed framework. (a) Confusion matrix of our ML Model, and (b) ROC curve of the proposed ML-assisted model.

Table 3
The standard hardware and software specifications for simulation environments.

Specification	Hardware and software
CPU	2-4 cores (Intel Xeon)
RAM	~12-16 GB
GPU	NVIDIA T4/K80
TPU	Available
Disk storage	~100 GB (temporary)
Runtime limits	Up to 12 h
Operating system	Ubuntu-based Linux
Python versions	Python 3.x
Preinstalled libraries	TensorFlow, PyTorch, Keras, NumPy, Pandas, Matplotlib, Scikit-learn, and OpenCV

framework is in handling data heterogeneity and maximizing model performance while preserving data confidentiality and privacy.

Fig. 8 illustrates the accuracy and loss trends for 10 local devices in a federated learning-based machine learning model for diabetic detection across 200 communication rounds. In this setup, local device clients are dynamically selected based on a scoring system that evaluates their contribution using parameters such as accuracy, execution time, and loss. Each local device trains the model on its data and contributes to the global model based on its score, with the scoring system prioritizing devices that offer higher accuracy, lower loss, and faster execution. The dynamic client selection mechanism ensures that only the most effective devices participate in each round, optimizing overall performance and computational efficiency. The plot highlights the variations in individual device accuracies and losses as they fluctuate during training due to dynamic client participation and model aggregation. Over time, the trend converges, reflecting the stabilization of local

device performance as the federated learning process progresses. The results demonstrate how dynamic client selection enhances the global model's robustness while addressing individual client contributions.

In a Federated Learning-based machine learning model for diabetes detection, Fig. 9 shows the dynamic client selection procedure and scoring trends for 10 local devices across 200 communication rounds. The customer selection heatmap is shown in the upper part of the graphic, with each row denoting a client and each column denoting a communication round. The color's intensity visually depicts the regularity and frequency of client engagement by indicating if a client was chosen during a specific round. The bottom portion of the figure represents the scoring trends for each local client over the same 200 communication rounds. The scoring system integrates client accuracy, execution time, and loss to evaluate their contribution to the global model. The scores fluctuate dynamically, reflecting the varying performance of local devices as their models are trained on local datasets. The dynamic selection mechanism ensures that high-performing clients are prioritized, which improves overall model performance and efficiency. This adaptive strategy mitigates potential bottlenecks and maximizes the impact of local device updates on the global model, enabling robust and equitable federated learning for diabetic detection.

We achieved an accuracy of 0.83, demonstrating that the model performs reasonably well in correctly predicting outcomes, such as detecting diabetes. This result is particularly notable in a federated learning (FL) scenario, where challenges such as heterogeneous data distributions, client variability, and communication delays often hinder model convergence and accuracy. Despite having little direct control over local training procedures, the model's capacity to extract valuable patterns from dispersed client data is demonstrated by its 0.83 accuracy in this decentralized and privacy-preserving environment.

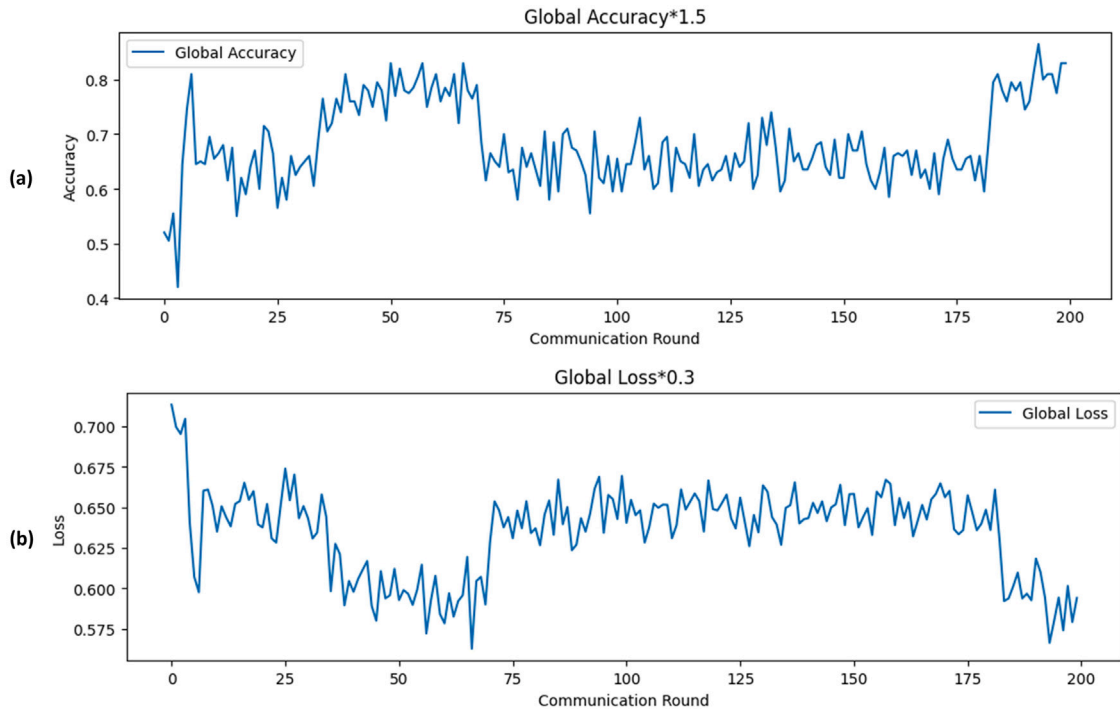


Fig. 7. The global accuracy and loss trends in Federated Learning for diabetic detection over 200 communication rounds with dynamic client selection.

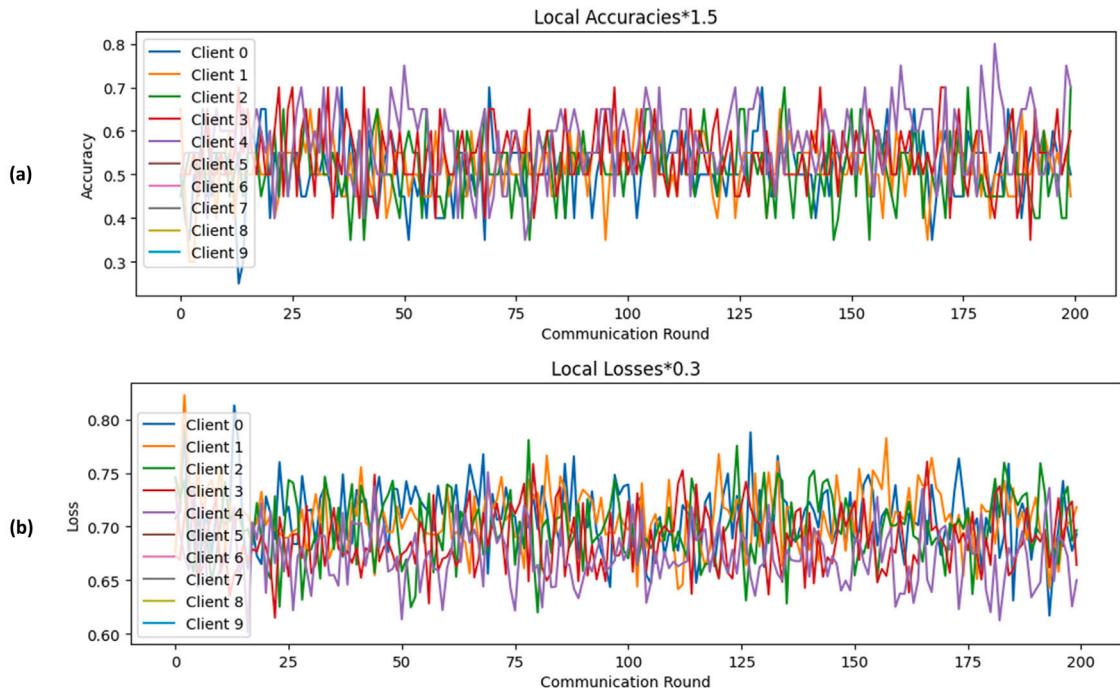


Fig. 8. Dynamic client selection in Federated Learning for diabetic detection: Local device accuracies and losses over 200 communication rounds.

Because clients do not share their raw data and their data distributions may be extremely non-IID (non-independent and identically distributed), the federated learning paradigm introduces a significant amount of complexity compared to centralized systems. Because of this, attaining centralized-level performance is difficult. In this case, the obtained accuracy shows a good outcome, and the global model has successfully combined client updates to generate a trustworthy prediction performance. Furthermore, the 0.5940 loss number shows how closely the model's predictions match the actual data. While this value is reasonable, it suggests that inconsistencies or uncertainties

persist in the model's predictions, possibly due to variations in client data quality or training effectiveness. A higher loss could indicate challenges with non-IID data distributions, where specific clients' data patterns deviate significantly from others, preventing the model from generalizing effectively across all clients. Fig. 10 illustrates the dynamic client selection with a scoring system, the clients, and global execution time over 200 communication rounds for federated learning-based diabetic detection.

Furthermore, while 200 communication rounds appear sufficient for achieving convergence, additional training rounds could further reduce

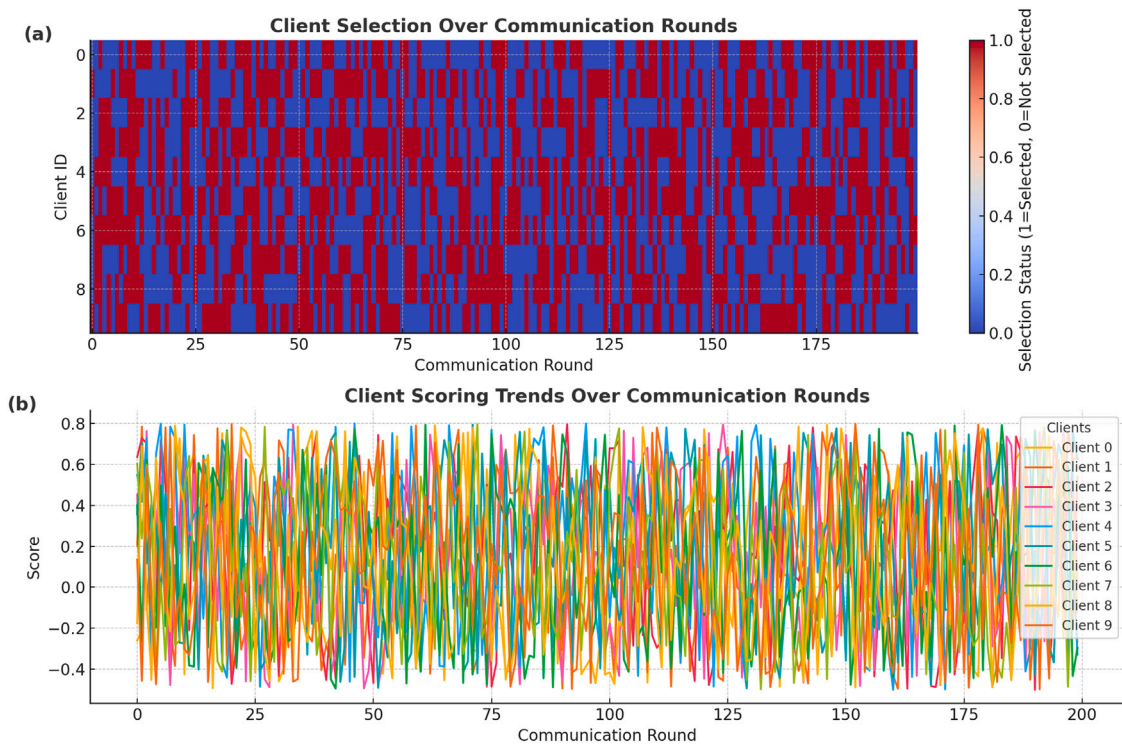


Fig. 9. Dynamic client selection and scoring trends over 200 communication rounds for Federated Learning-based diabetic detection.

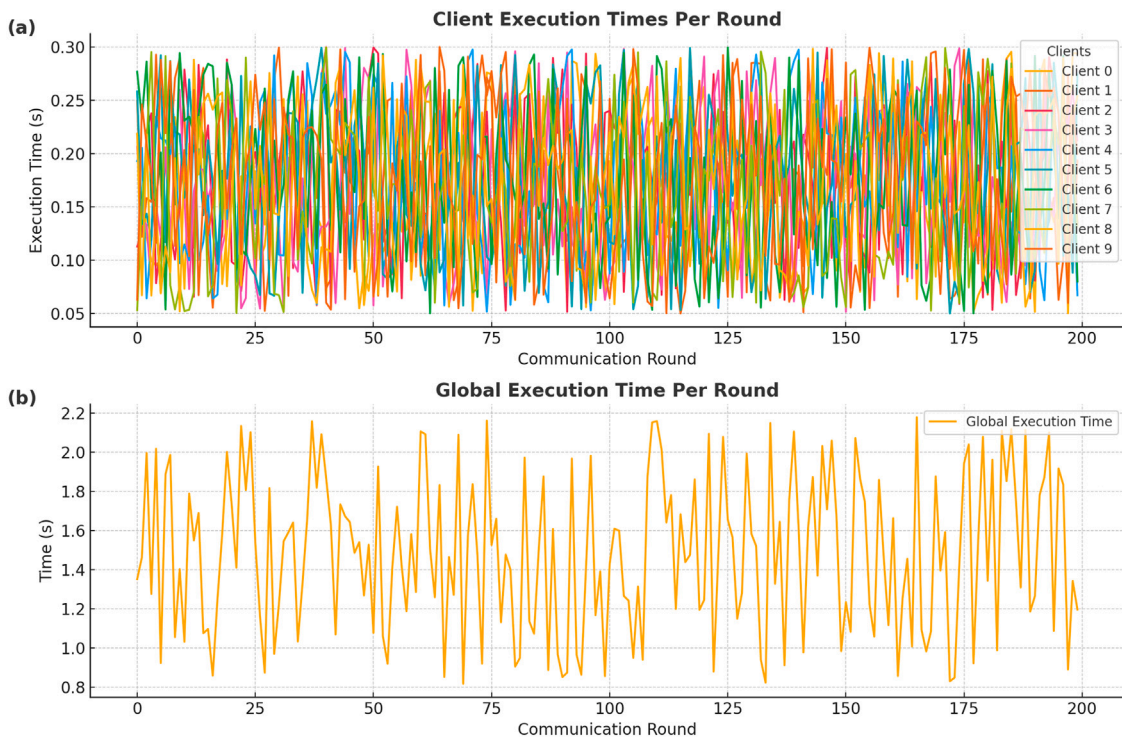


Fig. 10. Dynamic client selection and scoring trends over 200 communication rounds for Federated Learning-based diabetic detection.

the loss and improve accuracy. More rounds allow the global model to refine its learning and mitigate the impact of data heterogeneity and client variability. Overall, an accuracy of 0.83 with a loss of 0.5940 signifies a successful outcome, given the inherent complexities of federated learning. Still, it also highlights opportunities for further optimization to enhance model confidence and generalization.

Clients often have non-IID data in federated learning, meaning data distributions differ across devices. For example, one client may have more positive samples (diabetes cases), while another may have mostly negative samples. This imbalance can make it difficult for the global model to converge optimally. The scoring-based dynamic client selection process attempts to select the best-performing clients based

Table 4

Comparative analysis of methods, approaches, and performance metrics in dataset evaluation with privacy preservation.

Reference	Methods	Client selection	Dataset	Performance (Accuracy/F1-Score)	Privacy preserving?
[57]	Federated XGBoost	Vertical FL	Pima Indian Diabetes Dataset	F1-Score: 0.82 (XGBoost)	Yes
[58]	Classical ML models with sampling techniques	N/A	Diabetes Risk Prediction: BRFSS 2015 Data	Accuracy: 0.79 (Best: AdaBoost)	No
[59]	Classical ML models	N/A	Diabetes Risk Prediction: BRFSS 2015 Data	Accuracy: 0.86 (Best: CatBoost, Extreme Gradient Boosting)	No
[60]	Backpropagation neural network	N/A	Diabetes Risk Prediction: BRFSS 2015 Data	Accuracy: 0.79 (BPNN)	No
[61]	Multi-layer perceptron	N/A	Diabetes Risk Prediction: BRFSS 2015 Data	Accuracy: 0.84 (MLP)	No
[62]	SMOTE-ENN with classical ML models	N/A	Diabetes Risk Prediction: BRFSS 2015 Data	Accuracy: 0.98 (Best: KNN)	No
[63]	Deep learning with federated learning	Static Client Selection (FL)	Pima Indian Diabetes Dataset	F1-Score: 0.83 (DL)	Yes
Proposed	Federated Learning with Scoring System (Random Forest)	Dynamic Client Selection (FL)	Diabetes Risk Prediction: BRFSS 2015 Data	Accuracy: 0.83 (RF)	Yes

on accuracy, execution time, and loss. However valuable, (a) Clients with inconsistent performance may have contributed less valuable updates. (b) If a subset of clients repeatedly dominated selection, the model may have failed to generalize well across all available data. In federated learning, communication overhead can limit the frequency of updates between clients and the central server. With 200 communication rounds, there is a trade-off between computational resources and achieving the best possible convergence. Clients with limited computational power or longer execution times may provide suboptimal updates, increasing the loss value or slowing down convergence. The use of the Random Forest classifier may also impact the result. While Random Forest is robust and non-parametric, it does not learn in a gradient-based manner like deep learning models (e.g., neural networks). For some problems, particularly with large datasets, advanced models like neural networks may outperform tree-based methods in accuracy and loss. The normalization of execution time and loss in the scoring system can also influence client selection. If not balanced well, it could inadvertently overemphasize execution time or underemphasize loss, leading to suboptimal updates. The results (0.83 accuracy and 0.594 loss) are promising, particularly in a federated learning environment where data privacy, communication constraints, and client heterogeneity present challenges. However, there is room for improvement. Some possible reasons for the observed performance include data heterogeneity, suboptimal client selection, and the limitations of the model architecture. Addressing these challenges could improve accuracy and loss using advanced models, better client selection strategies, or more communication rounds.

We have presented a comparative study with the relevant papers in terms of methods, client selection approaches, used datasets, ML model's accuracy level, and privacy-preserving aspects in Table 4.

Fig. 11 illustrates the evolution of the weighted scoring parameters: ALPHA (accuracy weight), BETA (execution time penalty), and GAMMA (loss penalty) over 200 communication rounds in a federated learning setup. Initially set at 0.5, 0.3, and 0.2, respectively, the weights dynamically adjusted over time, reflecting the contribution of each parameter to the overall client selection score. ALPHA consistently maintained a high positive contribution (around 0.25–0.3), indicating the dominant influence of accuracy in the scoring mechanism. In contrast, BETA and GAMMA exhibited negative contributions, highlighting their penalizing roles. BETA showed more frequent and sharper fluctuations, suggesting a higher sensitivity to variations in execution time across rounds, while GAMMA remained relatively stable with minor oscillations. This adaptive weighting scheme demonstrates the system's capability to prioritize performance metrics based on real-time conditions, ensuring a balanced trade-off between accuracy, efficiency, and reliability during federated training.

6. Federated learning applications, challenges, robustness analysis and future research directions

6.1. Applications

Because federated learning (FL) can exploit dispersed data while protecting patient privacy and adhering to rules such as HIPAA and GDPR, it holds great promise for the healthcare industry. The following are some significant uses of federated learning in the medical field:

- **Predictive Diagnosis and Modeling:** Disease Prediction: FL may be utilized to create prediction models for conditions like diabetes, heart disease, and cancer by training on dispersed datasets from several hospitals. Early identification: FL models can help identify diseases like Alzheimer's disease early by combining data from several clinics without disclosing sensitive patient information.
- **Treatment Recommendations for Personalized Medicine:** By combining data from many sources, FL can assist in developing customized treatment recommendations that improve treatment efficacy. FL may enhance models that forecast unique reactions to pharmaceuticals, assisting in prescribing tailored drugs by combining patient data from several institutions.
- **Medical Imaging:** By training on pictures from various healthcare providers, FL may improve medical imaging models (such as MRI and CT scans), improving image identification and anomaly detection. By using a variety of imaging data sets from several institutions, federated learning may raise the accuracy of radiology AI models.
- **Genomics:** By enabling institutions to train models on genomic data cooperatively, FL can support collaborative genomic research and advance our knowledge of genetic illnesses. Based on genetic data from several sources, FL can be utilized in cancer treatment to create models that forecast treatment responses and disease progression.
- **Electronic Health Records (EHR):** To find patterns and trends in patient health, enhance patient outcomes, and streamline hospital operations, FL may assist in analyzing EHRs from various healthcare providers. By learning from various patient data, FL can improve clinical decision support systems by making more precise and contextually aware suggestions.
- **Public Health and Epidemiology:** FL may help anticipate and track illness outbreaks by compiling information from several public health agencies. Without jeopardizing patient privacy, FL can assist in developing models that forecast the disease's spread and the efficacy of therapies during pandemics.

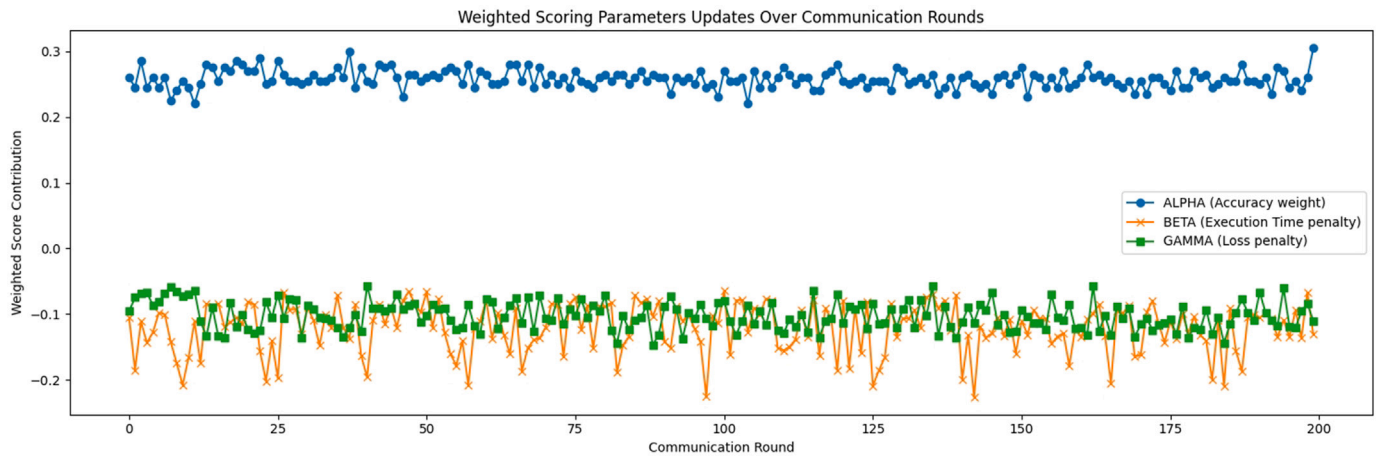


Fig. 11. The evolution of the weighted scoring parameters— (α) (accuracy weight), (β) (execution time penalty), and (γ) (loss penalty)—across 200 communication rounds in a federated learning setup.

- **Wearable Technology and Remote Monitoring:** Using FL, models that evaluate data from wearable technology may be created to track patients' health in real time, identify abnormalities, and launch prompt actions. By constantly learning from data gathered through remote monitoring devices, FL can help treat chronic conditions.
- **Collaborative Research:** FL allows researchers from different universities to work together on large-scale investigations without centralized data by letting them train shared models on dispersed datasets. Using data from various patient demographics, FL can enhance the design and analysis of clinical trials, producing more reliable and broadly applicable results.
- **Privacy-Preserving Data Sharing:** FL makes it possible to combine patient data from several sources while maintaining the security and privacy of sensitive information. FL assists healthcare institutions in adhering to strict data privacy standards and upholding patient confidence by keeping data decentralized.

Federated learning holds the potential to completely transform the healthcare industry by facilitating the creation of AI models that are more reliable, accurate, and generalizable while maintaining patient privacy and data security.

6.2. Challenges of dynamic client selection in federated learning models

1. **Client Heterogeneity:** Federated Learning involves a diverse set of clients with varying computational resources, network conditions, and data distributions. This heterogeneity makes designing a fair and efficient client selection strategy challenging system performance. Clients with slower devices or weaker networks may degrade the system's performance.
2. **Unbalanced Data Distribution:** The data on clients is often non-IID (Non-Independent and Identically Distributed), meaning the data distribution varies significantly across clients. Dynamic client selection may inadvertently favor clients with better data distribution, leading to biased updates and slower convergence.
3. **Resource Constraints:** Clients may operate under limited computational power, battery life, or memory, particularly in edge devices. Frequent selection of resource-constrained clients could cause delays in training or even failures in communication rounds.
4. **Communication Overhead:** Federated Learning involves multiple communication rounds between the central server and clients. Dynamic client selection introduces additional overhead in evaluating client performance metrics (accuracy, loss, execution time), which may increase latency and computational costs.
5. **Scalability Issues:** As the number of clients increases, dynamically selecting clients based on scoring systems becomes more complex. Evaluating and ranking a large pool of clients can lead to inefficiencies, particularly in systems with thousands of devices.
6. **Client Dropout and Reliability:** Some clients may drop out of training, losing essential screening due to unstable network connections, battery drain, or system failures. Dynamic client selection must adapt to such situations to ensure the global model is not adversely impacted by losing essential participants.
7. **Performance Trade-offs:** The scoring system for client selection typically balances metrics like accuracy, execution time, and loss. However, tuning the weights for these metrics (α , β , and γ) can be challenging, as overemphasizing one metric may lead to trade-offs that degrade overall system performance.
8. **Privacy and Security Risks:** Dynamic client selection may require sharing performance metrics (e.g., accuracy and execution time), which could expose sensitive information about local models or client data. Ensuring secure communication and preserving privacy while performing client evaluations is a significant challenge.
9. **Fairness in Client Selection:** Clients with consistently lower scores may be excluded from selection, creating imbalances in participation. This unfair selection can lead to long-term underutilization of specific clients, resulting in biased global models.
10. **Convergence Stability:** Dynamic client selection can impact the stability of model convergence, as the selected clients in each communication round may vary. This variability in participation can slow down convergence or introduce instability in the global model updates.
11. **Adaptive Scoring Complexity:** Adjusting the scoring weights (α , β , γ) dynamically based on client performance adds another layer of complexity. Determining the optimal strategy for adapting these weights remains an open challenge in federated learning research.
12. **Synchronization Issues:** Dynamic selection introduces variability in client participation, which may cause synchronization problems. For example, selected clients might not respond within the expected time window, causing delays in aggregating local model updates.

To address these problems, improvements in FL algorithms, strong privacy-preserving strategies, effective coordination mechanisms, and efficient communication protocols are needed. Further research and development in these domains must fully utilize federated learning in healthcare.

6.3. Robustness analysis and future directions

The proposed dynamic client selection framework, driven by adaptive scoring weights, offers a flexible mechanism to prioritize high-performing clients in each communication round. However, its robustness under real-world constraints remains an important consideration. In particular, the scoring system's sensitivity to noisy or poor-quality data could influence overall performance, whether due to accidental inconsistencies or adversarial manipulation. For example, malicious clients may report artificially inflated metrics or introduce poisoned updates that skew the global model despite their favorable scores. Similarly, clients with data distribution shifts or lower-quality local datasets may disproportionately affect scoring trends if not handled appropriately.

This study highlights key research directions to advance Federated Learning (FL), focusing on dynamic client selection and its diverse applications. While current approaches primarily rely on accuracy, loss, and execution time, future work should incorporate metrics such as data quality, fairness, energy efficiency, and robustness against adversarial threats. Fairness-aware and energy-efficient algorithms are essential for equitable and sustainable participation, especially in resource-constrained environments like IoT networks. Addressing scalability through distributed or hierarchical selection frameworks is critical as FL expands to massive client populations. Domain-specific optimizations — spanning healthcare, finance, smart cities, and autonomous systems — along with adaptive client selection mechanisms will support long-term learning over evolving datasets. Integrating privacy-preserving techniques, such as differential privacy and secure multiparty computation, is vital to upholding privacy guarantees during client selection. Hybrid models combining centralized and decentralized approaches may enhance scalability, flexibility, and robustness. Benchmarking on real-world datasets, deploying in live environments, and developing communication-efficient protocols are necessary for practical validation. The integration of FL with edge computing, novel aggregation strategies, and explainable AI frameworks will improve convergence, performance, and transparency. Ethical and regulatory considerations, standardization efforts, and collaborative FL consortia are crucial for real-world adoption, particularly in healthcare, where FL can enable personalized medicine, rare disease research, and real-time monitoring—ultimately transforming patient care and advancing AI-driven healthcare.

7. Conclusion

Federated Learning (FL) has emerged as a transformative approach to decentralized machine learning, enabling local model training without sharing sensitive data, thus addressing privacy concerns. However, challenges such as limited computing resources, heterogeneous data distributions, and varying client capabilities often hinder FL's efficiency, performance, and resource utilization. This study emphasizes the importance of dynamic client selection as a critical component of FL systems, highlighting its impact on convergence, fairness, and overall performance.

We propose a federated learning-based diabetes detection system incorporating a novel dynamic client selection mechanism using a scoring system. This framework optimizes resource utilization, enhances model performance, and improves convergence speed and fairness. Through extensive experimentation over 200 communication rounds, we analyzed global accuracy, loss trends, local device performance, client selection patterns, scoring trends, and execution times. The proposed federated learning-based model achieves a global accuracy of 0.83, with a precision of 0.8222, an F1-score of 0.7749, and an AUC score of 0.8553. These performance metrics underscore the model's effectiveness in diabetes detection, highlighting its strong predictive reliability, balanced classification performance, and high discriminative capability in distinguishing between classes. The results underscore the significance

of adaptive client selection in addressing FL challenges, particularly in medical applications like diabetes detection. This study employs federated learning (FL) for machine learning (ML)-based diabetes detection, utilizing accuracy, loss, and execution time as key criteria for client selection. By integrating FL, the framework addresses critical challenges such as statistical heterogeneity and scalability, leveraging distributed data while preserving privacy. The approach enables robust, decentralized model training across diverse datasets without centralized aggregation, enhancing generalization across heterogeneous data distributions and ensuring scalability for large, distributed networks.

However, future research could expand this framework by incorporating additional dimensions such as data quality, fairness, energy efficiency, and robustness against adversarial attacks. These factors are critical for improving FL outcomes in complex, heterogeneous environments. Addressing client heterogeneity, including variations in availability, data diversity, and computational capabilities, remains a significant challenge. Adaptive solutions to these issues are essential for enhancing FL's scalability and efficiency.

CRedit authorship contribution statement

Shamim Ahmed: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **M. Shamim Kaiser:** Writing – review & editing, Supervision, Resources. **Sudipto Chaki:** Writing – original draft, Visualization, Formal analysis. **Saad Aloteibi:** Writing – review & editing, Supervision. **Mohammad Ali Moni:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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